Li “Harry” Zhang

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Last updated: Oct 2026

RESEARCH INTERESTS

Artificial Intelligence, Machine Learning, Natural Language Processing
Large Language Models, Planning, Reasoning, Agents, Formal Methods, etc.

ACADEMIC AFFILIATIONS

Drexel University, Philadelphia, PA — Dec 2024 – Present
Tenure-Track Assistant Professor

University of Pennsylvania, Philadelphia, PA — Aug 2019 – Aug 2024
Ph.D. Computer and Information Science — GPA: 3.96/4.00
Adviser: Prof. Chris Callison-Burch
Thesis: *Structured Event Reasoning with Large Language Models*
Committee: Prof. Dan Roth (chair), Prof. Rada Mihalcea, Prof. Graham Neubig, Prof. Mark Yatskar, Dr. Marianna Apidianaki

University of Michigan, Ann Arbor, MI — Aug 2015 – Dec 2018
B.S.E. Computer Science, summa cum laude — GPA: 3.82/4.00
Mentors: Prof. Rada Mihalcea and Prof. Dragomir Radev

PUBLICATIONS

36 papers peer-reviewed and published in conferences and workshops
⊃ 13 first-authored ∪ 9 last-authored
Total citations: 4000+; h-index: 15+
(*Equal contribution; ^Mentored students)

[45] Theory-Scale Auto-Formalization of Logics for Computer Science; Yuming Feng, Frederick Pu, One An, Osbert Bastani, **Li Zhang**, Jiani Huang, Xujie Si and Ziyang Li; in NeurIPS 2026.

[44] Do Value Vectors in Deep Layers Need Context from the Residual Stream?; Muyu He, Yuchen Liu, Qingya Huang and **Li Zhang**; in EMNLP 2026.

[43] Robust Asynchronous Planning via Auto-Formalization; Jiayi Zhang*, Jianing Yin*, Ben Zhou and **Li Zhang**; in submission.

[42] Language Model Planners do not Scale, but do Formalizers? Owen Jiang, Cassie Huang, Ashish Sabharwal and **Li Zhang**; in submission.

[41] C. Huang, S. Mohan, Z. Yang, S. Tellex and **L. Zhang**. Language Model as Planner and Formalizer under Constraints. In ACL 2026.

[40] V. Menon, A. Cherney, E. Cloude, **L. Zhang**, T. Do. Evaluating the Impact of LLM-guided Reflection on Learning Outcomes with Interactive AI-Generated Educational Podcasts. In AIME-Con 2025.

[39] C. E. Kayan and **L. Zhang**. Prototype-Based Dynamic Steering for Large Language Models; in EMNLP 2026.

[38] M. He, Y. Zheng, Y. Liu, Z. An, B. Cai, J. Huang, L. Zhou, F. Liu, Z. Li and **L. Zhang**. Vision Language Models Cannot Plan, but Can They Formalize? In ECCV 2026 Workshop WMEAI; Best Paper Award.

[37] R. Wang and **L. Zhang**. Documentation Retrieval Improves Planning Language Generation. In AACL 2025.

[36] L. Gong, W. Zhu, J. Thomason and **L. Zhang**. Zero-Shot Iterative Formalization and Planning in Partially Observable Environments. In Findings of ACL 2026.

[35] P. Kagitha and **L. Zhang**. Addressing the Challenges of Planning Language Generation. In Findings of ACL 2026.

[34] R. Amonkar, M. Lai, R. Le Bras and **L. Zhang**. Are LLMs Better Formalizers than Solvers on Complex Problems? In the 2nd Workshop on Efficient Reasoning at COLM 2026.

[33] Y. Yuan, M. He, A. Shahid, J. Huang, Z. Li, **L. Zhang**. TurnaboutLLM: A Deductive Reasoning Benchmark from Detective Games. In EMNLP 2025.

[32] Not that Groove: Zero-Shot Symbolic Music Editing; Li Zhang; in submission to ISMIR.

[31] W. Hu, J. Duan, C. Wei, **L. Zhang**, Y. Zhang and K. Xu. DynaCode: A Dynamic Complexity-Aware Code Benchmark for Evaluating Large Language Models in Code Generation. In Findings of ACL 2025.

[30] C. Huang and **L. Zhang**. On the Limit of Language Models as Planning Formalizers. In ACL 2025.

[29] **L. Zhang**, P. Jansen, P. Clark, C. Callison-Burch and N. Tandon. PDDLEGO: Iterative Planning in Textual Environments. In *SEM 2024.

[28] T. Zhang*^, **L. Zhang***, Z. Hou^, Z. Wang^, Y. Gu, P. Clark, C. Callison-Burch and N. Tandon. PROC2PDDL: Open-Domain Planning Representations from Texts. In the 2nd Natural Language Reasoning and Structured Explanations Workshop at ACL 2024.

[27] Q. Lyu, K. Shridhar, C. Malaviya, **L. Zhang**, Y. Elazar, N. Tandon, M. Apidianaki, M. Sachan and C. Callison-Burch. Calibrating Large Language Models with Sample Consistency. In AACL 2023; **Area Chair Award**.

[26] Y. Lal, **L. Zhang**, F. Brahman, B. Majumder, Peter Clark and N. Tandon. One Size Does Not Fit All: Customizing Open-Domain Procedures. In Findings of ACL 2024.

[25] B. Majumder, B. Dalvi, P. Jansen, O. Tafjord, N. Tandon, **L. Zhang** and C. Callison-Burch, Peter Clark. CLIN: A Continually Learning Language Agent for Rapid Task Adaptation and Generalization. In COLM 2024.

[24] Z. Hou^, **L. Zhang** and C. Callison-Burch. *Choice-75: A Dataset on Decision Branching in Script Learning*. In LREC-COLING 2024.

[23] **L. Zhang**, H. Xu^, A. Kommula, N. Tandon and C. Callison-Burch. *OpenPI2.0: An Improved Dataset for Entity Tracking in Texts*. In EACL 2024.

[22] **L. Zhang***, L. Dugan*, H. Xu*^ and C. Callison-Burch. *Exploring the Curious Case of Code Prompts*. In preprint. In the 1st Natural Language Reasoning and Structured Explanations Workshop at ACL 2023.

[21] T. Zhang^, I. Tham, Z. Hou^, Jia. Ren, L. Zhou, H. Xu^, **L. Zhang**, L. Martin, R. Dror, S. Li, H. Ji, M. Palmer, S. Brown, R. Suchocki, C. Callison-Burch. *Human-in-the-Loop Schema Induction*. In preprint; in ACL 2023 Demos.

[20] Q. Lyu*, S. Havaldar*, A. Stein*, **L. Zhang**, D. Rao, E. Wong, M. Apidianaki and C. Callison-Burch. *Faithful Chain of Thought Reasoning*. In IJCNLP-AACL 2023.

[19] **L. Zhang***, H. Xu*^, Y. Yang, S. Zhou, W. You, M. Arora and C. Callison-Burch. *Causal Reasoning of Entities and Events in Procedural Texts*. In Findings of EACL 2023.

[18] **L. Zhang** and C. Callison-Burch. *Language Models are Drummers: Drum Composition with Natural Language Pre-Training*. In 1st Workshop on Creative AI across Modalities at AAAI 2023.

[17] Y. M. Cho^, **L. Zhang** and C. Callison-Burch. *Unsupervised Entity Linking with Guided Summarization and Multiple Choice Selection*. In EMNLP 2022.

[16] S. Gehrmann, …, **L. Zhang**, …, H. Zhu, S. Brahma, Y. Li, ... *GEMv2: Multilingual NLG Benchmarking in a Single Line of Code*. In EMNLP 2022.

[15] A. Srivastava, …, **L. Zhang**, Q. Lyu and C. Callison-Burch, … *Beyond the Imitation Game: Quantifying and extrapolating the capabilities of language models.* In TMLR.

[12] Q. Lyu, H. Zheng, D. Li, **L. Zhang**, M. Apidianaki, and C. Callison-Burch. *Is "my favorite new movie" my favorite movie? Probing the Understanding of Recursive Noun Phrases*. In NAACL 2022.

[11] **L. Zhang**, I. Jindal and Y. Li. *Label Definitions Improve Semantic Role Labeling*. In NAACL 2022.

[10] **L. Zhang***, S. Zhou*, Q. Lyu, Y. Yang, G. Neubig and C. Callison-Burch. *Show Me More Details: Discovering Event Hierarchies from WikiHow*. In ACL 2022.

[9] Y. Yang, A. Panagopoulou, Q. Lyu, **L. Zhang**, M. Yatskar and C. Callison-Burch. *Visual Goal-Step Inference using wikiHow*. In EMNLP 2021; presented at the 2nd Workshop on Advances in Language and Vision Research at NAACL 2021.

[8] **L. Zhang***, Q. Lyu* and C. Callison-Burch. *Goal-Oriented Script Construction*. In INLG 2021.

[7] **L. Zhang**, Q. Lyu and C. Callison-Burch. *Intent Detection with WikiHow*. In AACL-IJCNLP 2020.

[6] **L. Zhang***, Q. Lyu* and C. Callison-Burch. *Reasoning about Goals, Steps, and Temporal Ordering with WikiHow*. In EMNLP 2020; presented at Workshop on Enormous Language Models at ICLR 2021.

[5] **L. Zhang**, H. Zhu, S. Brahma and Y. Li. *Small but Mighty: New Benchmarks for Split and Rephrase*. In EMNLP 2020.

[4] **L. Zhang**, S. R. Wilson and R. Mihalcea. *Multi-Label Transfer Learning for Semantic Similarity*. In *SEM 2019 and presented at NAACL 2019.

[1] C. Finegan-Dollak, J. K. Kummerfeld, **L. Zhang**, K. R. D. Ramanathan, S. Sadasivam, R. Zhang and D. Radev. *Improving Text-to-SQL Evaluation Methodology.* In ACL 2018.

External Funding

Decision-Embedded Deep Learning for Transit Systems ($432,572) 2024 - 2027

NSF

• Inherited from Dr. Kaidi Xu.

Alexa Prize TaskBot Challenge ($250,000) 2021 - 2022

Amazon

• Primarily authored, applied, and received a stipend award of $250,000 to lead University of Pennsylvania's effort in the Alexa Prize TaskBot Challenge 2021.

INDUSTRY EXPERIENCE

Research Intern Apr 2023 – Dec 2023

Allen Institute for Artificial Intelligence (AI2) Seattle, WA

Research Intern Apr 2019 – Jun 2019; May 2021 – Aug 2021

IBM Research San Jose, CA

Software Engineer Intern May 2017 – Aug 2017

Goldman Sachs Group, Inc. Jersey City, NJ

ACADEMIC SERVICE

Chair

• Area Chair:
EMNLP 2026, ACL2026, NAACL 2025, AACL 2025, EMNLP 2025, ACL 2025, ACL 2024, EMNLP 2024, COLING 2024

• Program Chair:
Mid-Atlantic Student Colloquium on Speech, Language and Learning (MASC-SLL) 2023

• Program Chair:
1st Workshop on Data Science with Human in the Loop at EMNLP 2022 2022

• Session Chair:
ACL 2026, AACL 2020

Reviewer

• Association of Computational Linguistics (ACL)	recurring
• North American Chapter of ACL (NAACL)	recurring
• Empirical Methods in Natural Language Processing (EMNLP)	recurring
• Association for the Advancement of Artificial Intelligence (AAAI)	recurring
• Conference on Language Modeling (COLM)	recurring
• International Conference on Language Resources and Evaluation (LREC)	recurring
• International Conference on Computational Linguistics (COLING)	recurring
• Computer Speech and Language (CSL) journal.	recurring

TEACHING

Instructor — Applied Natural Language Processing	Spring 2025-2027
CS T780: The graduate level NLP course	Drexel University
Instructor — Applied Machine Learning	Spring 2026
CS T614: The graduate level ML course	Drexel University
Teaching Assistant — Computational Linguistics	Jan 2020 – Dec 2020
CIS 530: The graduate level NLP course	University of Pennsylvania
Teaching Assistant — Natural Language Processing	Sept 2018 – Dec 2018
EECS 595: The graduate level NLP course	University of Michigan
Teaching Assistant — Programming and Data Structures	Sept 2016 – Apr 2017
EECS 280: An introductory programming course	University of Michigan
Tutor — Elementary Chemistry	Sept 2016 – Dec 2016
Science Learning Center	University of Michigan

ADVISING

PhD Students

Cassie Huang	Jan 2025 – present
Ceyhun Efe Kayan	Sep 2025 – present
Brian Tomasette	Sep 2026 – present

Master Students

Prabhu Prakash Kagitha	Feb 2025 – present

Undergraduate Students

Rikhil Amonkar	Feb 2025 – present
Stuti Mohan	Jun 2025 – present

Interns and Visiting Students

Owen Jiang, Jiayi Zhang, Jianing Yin, Lingze Zhang, Jen Shi, Kevin Nguyen

Alumni and Past Students

Bo Sun, Muyu He, Yuan Yuan, Renxiang Wang, Chimezie Maduno , Krystal Gong, Tianyi Zhang, Hainiu Xu, Zhaoyi Hou, Young-Min Cho

AWARD

- Featured in AAAI 2026 New Faculty Highlights